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RESEARCH ARTICLE

EC-YOLO: Effectual Detection Model for Steel Strip Surface Defects Based on YOLO-V5

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ABSTRACT Defect detection is extensively utilized within the metal industry, particularly for identifying surface imperfections on steel strips. However, the current methods still face challenges in detecting small and elongated defects on steel strips. Such defects occupy a relatively small pixel percentage within the entire image. The repeated downsampling in convolutional networks, coupled with the dynamic changes in the receptive field, can result in the potential loss of these minute defects. To mitigate the problem, our paper proposes EC-YOLO, a real-time defect detection network for steel strips of the above peculiar defects. Firstly, the 1D convolution in the efficient channel attention bottleneck (EB) module enhances the feature extraction ability of the backbone for small and elongated defects, while also facilitating the attentional mechanism for modeling channel features. Secondly, Context Transformation Networks integrate cross-stage localized blocks, referred to as CC modules, to enhance the understanding of feature semantic contextual information. Thirdly, a self-constructed dataset containing both small and elongated defects is used for understanding where such defects are more relevant in feature fusion and extraction. On the public datasets GC10-DET and NEU-DET, the improved model achieves mean Average Precision (mAP) scores of 71% and 83%, respectively, surpassing the performance of other mainstream models. The mAP of the enhanced model on the SLD-DET dataset reaches 87.5%, demonstrating its superiority in detecting both small and elongated defects.

INDEX TERMS Defect detection, small and elongated defects, attention, steel strips.

I. INTRODUCTION

More recently, object recognition has steadily evolved into one of the most intriguing projects in machine vision. Object detection can locate the target of interest in pictures and videos combined with completing tasks of location and classification. It is now extensively used in many domains, including machine detection, image retrieval, video surveillance, and automatic driving systems [1]. High-end hot-rolled strip products are extensively used in industry, household appliances, and construction. As the demand for high-end hot-rolled strip products increases in downstream sectors, the surface quality of the strip is becoming increasingly prominent [2]. Because the strip material in the rolling production process is affected by various aspects of the production process and equipment, the material surface is susceptible to inclusions, rolled-in scales, crazing, patches, scratches, creases, and other defects. These defects impair the corrosion resistance, wear resistance, and fatigue limit of the manufactured article, as well as other performance-related properties, potentially reducing the service life of the strip and leading to safety accidents [3]. Therefore, efficient detection of strip defects in industrial production is imperative. This paper aimed to utilize object detection to diagnose strip surface defects more efficiently, which will help the strip production industry separate defective products.

The evolutionary process of object recognition can generally be categorized into two distinct stages. The initial

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phase, which predates the year 2014, is commonly referred to as the conventional object detection phase. The conventional methods can be further subdivided into two stages. The first is target instance detection, including speeded-up robust features (SURF) [4], scale-invariant feature transform (SIFT) [5], and various improved algorithms. These methods usually use the correspondence between instance objects and templates to detect target instances. The second category, called category detection methods, includes Viola–Jones detection [6], [7], histogram of orientated gradients (HOG) [8], deformable parts model (DPM) [9], etc. These methods focus on picking regions that may contain detection objects in advance, performing feature extraction on these regions using different algorithms, and finally implementing detection and classification of the extracted features. The evolution of traditional detection algorithms is hindered by various problems, such as poor recognition, low accuracy, and greater reliance on manual labor [10].

After 2014, object detection entered the second period, called the deep learning object detection phase. During this period, detection approaches could be further categorized into two categories. The first one is the two-stage recognition algorithm represented by RCNNs (recurrent convolutional neural networks), including R-CNN [11], Fast R-CNN [12], and Faster R-CNN [13]. A two-stage object detection approach typically involves generating target bounding boxes that represent the location of target in advance, which can then be used for classification and regression. The second category is the one-stage recognition approach represented by the SSD (single-shot detector) series [14], [15], [16] and the YOLO (You Only Look Once) series [17], [18] of algorithms. A one-stage detection model directly selects the whole picture as inputs. It can achieve the prediction of the class and position of the bounding box in the detection head without predicting the target candidate boxes.

Because of the excellent performance of deep learning and CNNs in object detection, they have gradually become the leading research direction in defect detection. Researchers have proposed many algorithms with high accuracy and efficiency. Zhao et al. [19] added a squeeze-and-excitation (SE) attention mechanism on the SSD network called SE-Net, which improved the sample size imbalance in fabric detection. However, the dimension reduction operation in SE-Net inevitably entails some shortcomings. The approach fails to adequately capture the interdependencies among the channels, which is particularly detrimental when dealing with small and extended defects characterized by a minimal number of pixels. Li et al. [20] modified the YOLOv5 backbone using the module combined CNN with Transformer (TRANS) to alleviate the poor detection of minor defects and background interference in steel surface defect detection. The TRANS module leverages a transformer-style self-attention mechanism to improve global contextual feature extraction. Nevertheless, the self-attention approach used in this case is limited by its direct application to a 2D feature map, which results in isolated queries and keys in each spatial location for the attention matrix. Thus, the approach fails to capture the abundant contextual information that can be obtained from neighboring keys. Additionally, minor and elongated defects, which occupy fewer pixels, inevitably produce an aliasing effect during feature fusion. In other words, deep large defects may cover shallow small defects, and small defects may in turn blur the characteristics of large defects, resulting in feature dilution after fusion.

To address the above problems, the EB module promotes channel attention to small and elongated defects by non-decreasing 1D convolution, boosting backbone feature extraction for defect detection. Then, the CC block captures static and dynamic contexts using a 3×3 convolution and contextual self-attention, facilitating feature fusion and mitigating aliasing effects. This advancement allows for more accurate and holistic representations of complex data, enhancing the accuracy and reliability of data-driven decision-making processes. Furthermore, we explore the impact of attention utilization as well as the mechanisms through which small and elongated defects influence feature extraction and fusion.

The major contributions of our work are:

- To address the potential loss of small and long-range defect features due to multiple downsampling operations in the original YOLOv5 backbone, we incorporate the EB block within the backbone network to enhance the capture of dependencies between channels.
- Aiming at the YOLOv5 primitive neck network may exacerbate the aliasing effect when fusing small and elongated target scales, we introduce a static and dynamic joint self-attention CC block in the neck network to facilitate global contextual feature fusion.
- Aiming to address the scarcity of datasets containing small and elongated defects, we present a self-constructed dataset, SLD-DET. This dataset has been carefully curated to facilitate the analysis of unique defect characteristics and to enhance the detection of defects on steel surfaces.
- Aiming to improve the detection accuracy of small-sized and elongated defects, we have designed experiments to investigate whether these defects require more attention to semantic or graphical information during feature extraction and fusion.

II. RELATED WORK

With computer vision being thoroughly researched in the last few years, defect detection by computer has become a powerful assistant for surface inspection in the metal industry. Yet, up to now, several problems in defect detection on metal surfaces are intractable and constraining the development in this direction. i) The low-contrast between the defect and the ground. ii) In real industrial scenarios, the accuracy and speed of model detection are difficult to trade-off. iii) Large-scale variations in defect size, especially small-sized defects, are not susceptible to identification. For these difficulties, researchers have investigated the visual automatic detection techniques for steel surface defects.

A. STEEL SURFACE DEFECT DETECTION

Guan et al. [21] first trained the model using VGG19 to obtain feature images and then evaluated the image quality using a decision tree as well as structural similarity to categorize steel surface defects. Hatab et al. [22] used a single-stage target detector YOLO to detect steel surface defects, and the experiments proved that a fine-tuned YOLO detector could fulfill the steel surface defects detection task well. Deshpande et al. [23] utilized a siamese convolution neural network capable of sharing network weights to accomplish single sample recognition on steel surface defects. To eliminate the complicated defect background interference, Lu et al. [24] employed the channel space adaptive enhancement feature pyramid network (CA-FPN) to extract feature information of varying scales. Thereby amplifying the semantic information of the detected defects to the fullest extent possible, while simultaneously reducing background interference to a minimum. Chen et al. [25] improved the YOLOv3 model by utilizing MobileNetV2, an extended feature pyramid network (EFPN), and a feature fusion module. The lightweight backbone and multiscale feature fusion allow the model to achieve dual detection accuracy and speed. Li et al. [26] innovatively combined a Transformer encoder and a CNN to extract features related to surface defects of strip steel efficiently and combined with the traditional multilayer perceptron classification to boost strip steel surface defect detection capability.

In pursuit of focusing on objects of interest, the attention mechanism has emerged as a pivotal player in the integration of computer vision with attention mechanisms. Its widespread adoption has also impacted the domain of steel surface defect detection. Cheng et al. [27] integrated channel attention with RetinaNet in a CNN model, which significantly enhanced the effectiveness of the network in extracting feature maps. It indicates that the attention mechanism can also work for surface defect detection in steel. Zhao et al. [28] introduced the Multiple Feature-Maps Interaction Pyramid Network (MFIPNet), which incorporates an attention mechanism to enhance the capability of the network in adapting to the diverse complexities inherent in defect categories and their respective features. Its attention mechanism, used as a selector for multiple feature maps, unequivocally balances structural regularization and information, resulting in a superior ability of MFIPNet to discover surface defects on steel. Yeung and Lam [29] designed a fused-attention network (FANet) to detect surface flaws on different kinds of strips, which ensures immediate identification due to the selective application of the fused-attention framework to a single well-balanced feature map. Wang et al. [30] added weight matrices to steel defect feature maps by fusing attention mechanisms and adaptive deformable convolution for the superior feature. Wang et al. [31] utilized the spatial attention mechanism and multi-scale explore block to facilitate the steel surface defect recognition focusing on the defect information and exploring the surface defect size variations further. These methods dramatically improve target detection performance for complex surface defects.

B. DETECTION OF SMALL DEFECTS ON STEEL SURFACES

A convolutional neural network is a major tool for surface defect detection. Still, the constant stacking of convolution and downsampling operations will make the information of small defective targets continuously missing, especially steel surface defects with complex and diverse backgrounds, making defect detection more difficult.

Liu et al. [32] expanded the small defects to enlarge the number of minor defects and strengthen the accuracy of the detection task when facing the task of small defects on the steel surface. Akhyar et al. [33] proposed an upgraded RetinaNet defect detection module when confronted with the problem of small defects and class imbalance on steel surfaces, where the feature pyramid network and anchor point strategy improve the accuracy but cost more time. Zhao et al. [34] utilized MobileNetV3, a lightweight backbone network, and the TRANS module to efficiently extract contextual information from each level of images, obtaining a balance of speed and accuracy for detecting small defective targets on steel surfaces. Chen et al. [35] incorporated a feature fusion block to effectively amalgamate shallow and deep features from the backbone net, enhancing the identification of minor defects upon steel outsides. Zhong et al. [36] proposed the weighted matrix decomposition model (WMD) to identify small flaws on complex surfaces. The sparse moment and low rank are utilized to solve the sparse problem and low rank of small defects on steel surfaces and boost the dissimilarity between the defects and the ground. Zhang et al. [37] used YOLOv5 as a benchmark and added a layer of microscale detection to minimize the miss of feature information of minor target flaws on the steel surface. Yasir et al. [38] used a channel shuffle mechanism and depth-wise convolution to concentrate on extracting small target features of steel surfaces to increase the model characterization capability and feature propagation. Lu et al. [39] combined the CA-FPN and the CA-AutoAssign to boost the semantic information of minor defects on the steel surface and the adaptive fusion for multi-scale features to mitigate the interference of complicated background on defect recognition.

III. METHODOLOGY

A. THE BASELINE MODEL YOLOV5

The YOLO family of algorithms is a masterpiece of one-stage object detection, which was first proposed by Redmon et al. [17] in 2016. It has been broadly applied in industry and other fields through the development of YOLOv1, YOLOv2, and YOLOv3. The YOLOv5 algorithm is an upgraded edition of the YOLOv3 system. The YOLOv5 model architecture is shown in Figure 1, which can be approximately separated into four parts according to the model function: input preprocessing part, backbone feature

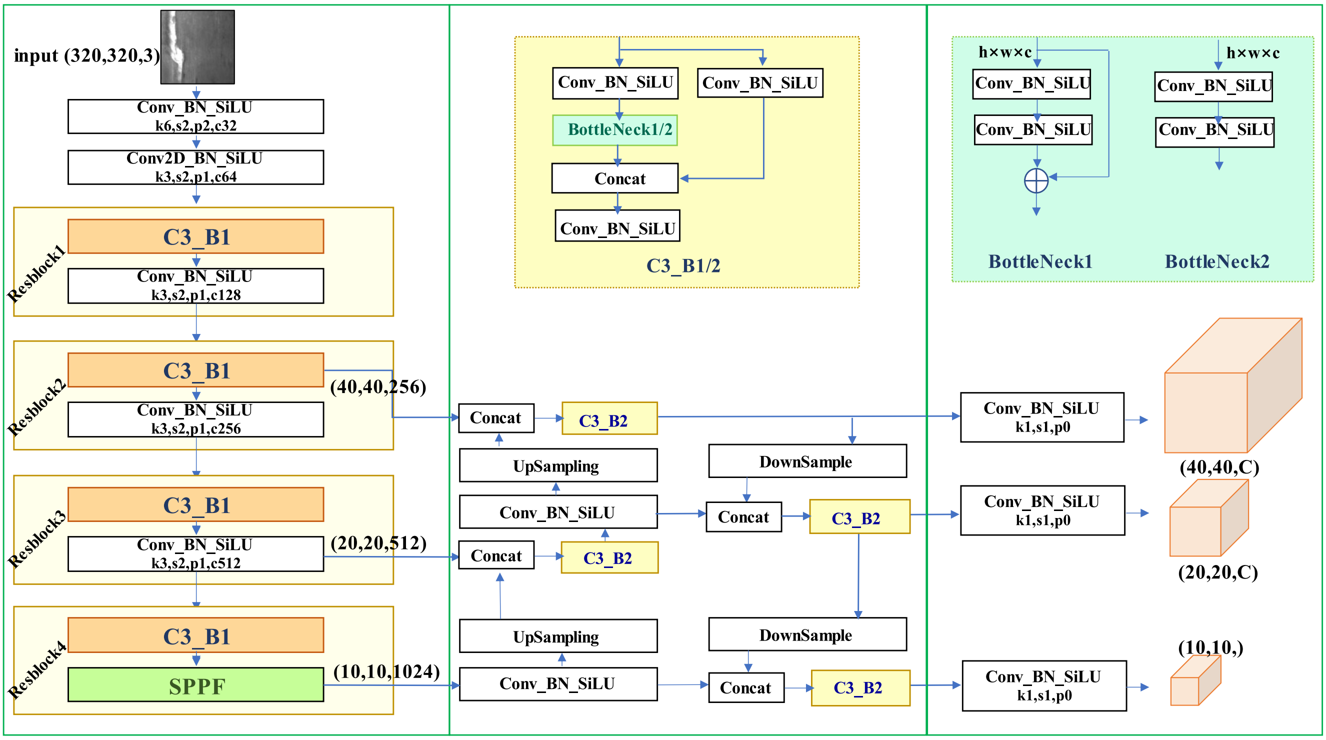


FIGURE 1. YOLOv5 overall architecture diagram.

extraction part, neck feature fusion part, and head prediction part. After inputting an image into the model, the YOLOv5 algorithm splits the picture into n × n cells. When an object centroid drops within a particular cell, the cell is answerable for forecasting the coordinates and categories of this target. In this way, the classification of the target detection is transformed into a regression problem. Recent developments have shown that the industry is more inclined to use a straightforward approach to facilitate detection models for better accuracy and detection rates.

TABLE 1. Comparison of the performance of YOLOv5 model versions in the NEU-DET dataset.

Models	mAP@0.5(%)	Params (M)
YOLOv5s	0.79	7.20
YOLOv5m	0.819	21.20
YOLOv5l	0.817	46.50
YOLOv5x	0.815	86.70

The latest YOLOv5 v6.0 model, with its ease of use, high speed, and accuracy, has been widely implemented in scientific research and manufacturing projects. The YOLOv5 algorithm is robust for recognizing special classes of defects. Due to its scalability, we chose this model as a benchmark for improving steel surface small and elongated defects detection accuracy. Upon the website GitHub, four versions of the YOLOv5 algorithm are made available to researchers: YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x, which increase in order with adjustments made to network width and depth. As shown in Table 1, the YOLOv5s parameters are three times, six times, and more than twelve times smaller than YOLOv5m, YOLOv5l, and YOLOv5x, respectively. However, the mean average precision (mAP) performance is only 2.9%, 2.7%, and 2.5% less than v5m, v5l, and v5x on the NEU-DET dataset, respectively. Considering our lightweight detection requirements, YOLOv5s v6.0 is adopted as the subsequent experimental model in this paper.

B. EFFICIENT CHANNEL ATTENTION (ECA)

Recently, attention mechanisms have performed remarkably well in convolutional networks, which have attracted the widespread attention of related research scholars. To process information in a computer vision module, the introduction of an attention mechanism can better align with the concept of visual attention patterns. More importantly, it can allocate limited computational resources to the region of interest based on the focus of attention. The Squeeze-and-Excitation Network (SENet) [44], introduced in 2018, is a representative network that incorporates attention mechanisms into convolutional modules. The method effectively leverages the interdependencies between information channels to extract the interchannel connections within each convolutional module, thereby enhancing the accuracy of the network model. As shown in Figure 2, the features input to SENet are pooled globally and averaged one by one according to the number of channels C. The consequent 1 × 1 × C features are then fully connected in two layers to capture the nonlinear interactions across channels, reducing the module dimensionality.

According to the experiments reported in [45], the addition of a fully connected layer to SENet enhances its performance under non-dimensional reduction conditions. However, this modification significantly increases the number of model parameters. The degradation operations employed by SENet in its fully connected layer may impede the capture of dependencies between channels, thereby negatively impacting network performance. The direct addition of full connectivity to the linear operation introduces numerous parameters to the network, compromising its lightweight nature and high accuracy.

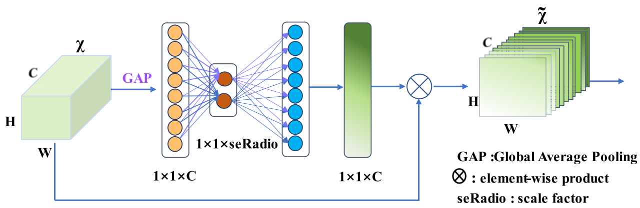


FIGURE 2. SE block.

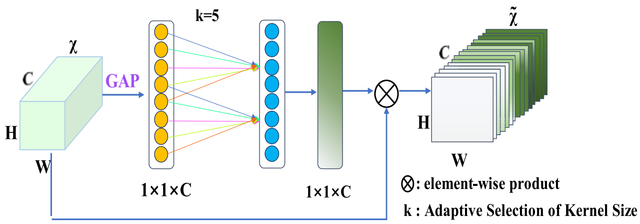


FIGURE 3. ECA block.

The ECA module [45] is displayed in Figure 3. After global averaging pooling, the channel weight calculation is done quickly by considering each channel feature and the k nearest neighbor features, using a one-dimensional convolution instead of channel downscaling in full connectivity. The parameter k represents the number of neighboring channels involved in the computation of a specific channel weight. This parameter significantly affects both the efficacy and computational efficiency of the ECA module. Concretely, k is directly correlated to the size of channel C. The large k can capture long-range dependencies between channel features, whereas a small k is more conducive to capturing short-range interactions. There is a nonlinear mapping relationship between the one-dimensional convolution k and channel C. The ECA module depicts this mapping relationship with an exponential function:

$$C = \Phi(k) = 2^{(\gamma \times k - b)}, \tag{1}$$

$$k = \psi(C) = \left| \frac{log_2(C)}{\gamma} + \frac{b}{\gamma} \right|_{odd}. \tag{2}$$

In equations (1) and (2), we make $\gamma = 2$, $b = 1$ for changing the proportion among the quantity of channels C and the size k sum of convolution kernel, and odd indicates the nearest odd number of k.

C. CONTEXTUAL TRANSFORMER NETWORK (CoT)

Owing to the accumulated knowledge and integration research directions in artificial intelligence, the transformer-style network architecture has gone from a single solution for natural language processing tasks to producing excellent results in computer vision. It has received an enormous concentration in the field. After introducing the object detection method with a transformer [45], it transformed the object detection problem into an ensemble prediction, simplifying the process of the target detection work and improving the drawbacks of needing to design the relevant components manually. However, existing transformer-style-based design tasks directly utilize a self-attention mechanism on 2D feature maps. A self-attention sketch is shown in Figure 4(a), where Q, K, and V in the traditional self-attention are obtained by a 1×1 convolution. The pairwise multiplication of Q and K ignores the rich contextual information embedded in the adjacent key. This poses a dilemma to the feature extraction task and weakens the ability of the self-attention mechanism to represent 2D feature maps. In 2021, the Jingdong AI Research Institute proposed the contextual transformer networks (CoT) block [46], which uses a self-attention mechanism in a transformer combined with convolutional operations to capture static and dynamic contextual information in the target image.

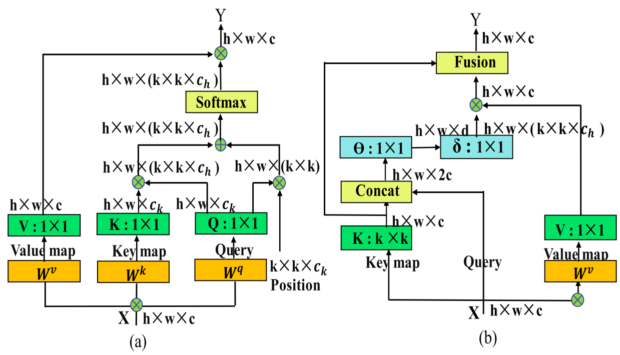


FIGURE 4. Self-attention(a) and CoT(b).

Compared with Figures 4(a) and 4(b), the CoT module in Figure 4(b) implements a 3×3 convolution of all adjacent keys in the 3×3 grid, and the obtained key features represent the static contextual information in the 3×3 range of the target image. Following that, the attention matrix of the module is obtained by using two 1×1 convolutions after stacking the keys and queries. Finally, the attention matrix is aggregated with values to get the input image with dynamic contextual description. Centralized extraction of static and dynamic information allows for more comprehensive feature fusion to mitigate aliasing. W^v, W^k, and W^q in Figure 4 represents three 1×1 embedded convolution, which are the parameters that the network needs to learn. Experiments [46] have shown that using the CoT block to substitute the 3×3 convolution in a standard convolutional architecture, such as ResNet, can effectively improve the model performance.

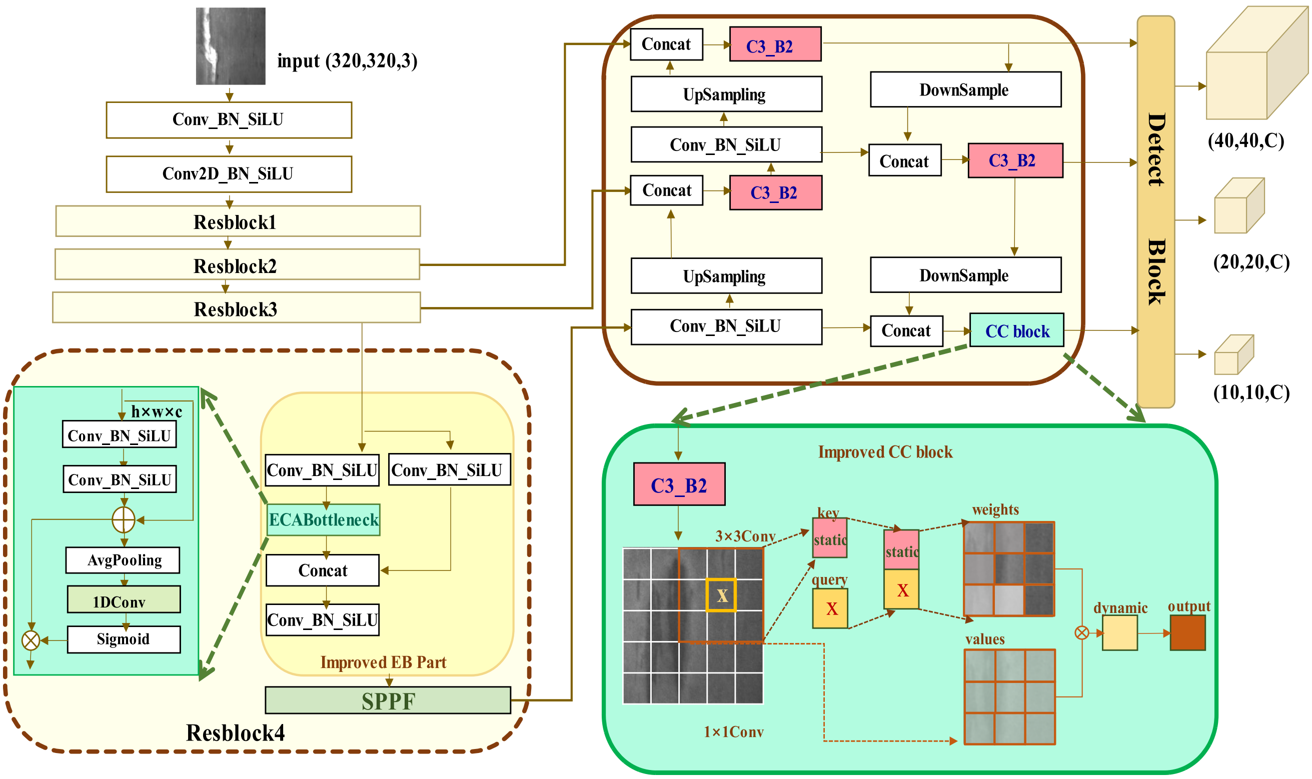


FIGURE 5. Proposed method.

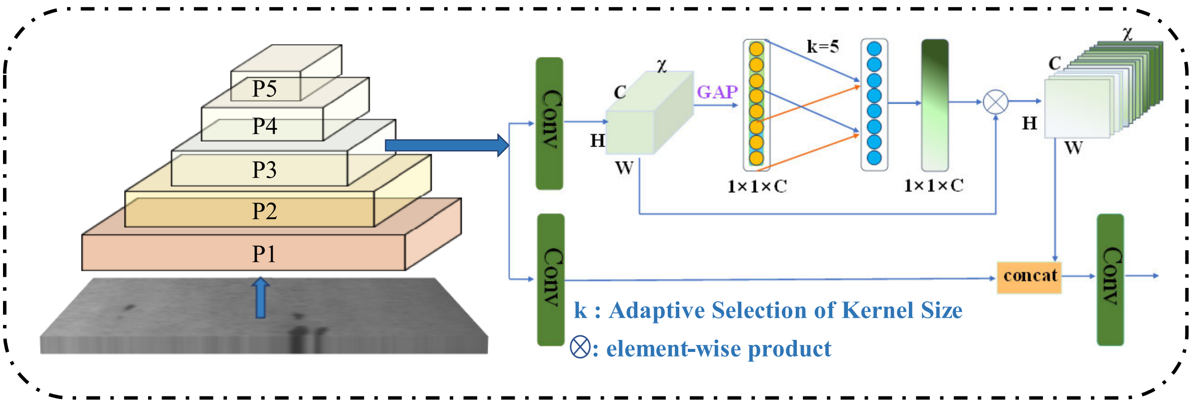


FIGURE 6. The backbone improvement sketch.

D. PROPOSED METHOD

Two prerequisites are essential when utilizing a defect detection model for decision-making in detecting defects. Firstly, the backbone network must be capable of extracting features from defects of varying sizes. Secondly, effectively fusing the extracted shallow and deep feature information poses a challenge. Specifically, inadequate pixel coverage of small and elongated defects results in insufficient attention, making it difficult to provide relevant information for diagnosis. Moreover, even if the backbone network can obtain ample multi-scale feature data, the fundamental FPN-based structure may cause an aliasing effect on small and elongated targets during feature fusion in the neck network. To address these issues, we have proposed an efficient basic building block (EB) that enhances feature perception ability by incorporating a channel attention mechanism. Additionally, we have modified the neck network PAFPN by integrating a CC block, which utilizes the self-attention of CoT to effectively combine contextual information from features of various sizes. The CC block emphasizes the extraction of shallow positional details while maintaining a balance with deep semantic features, thereby reducing the aliasing effect of the PAFPN. The combination of the EB and CC modules enables the enhanced model to maximize the use of defect information to fulfill the requirements for inspection.

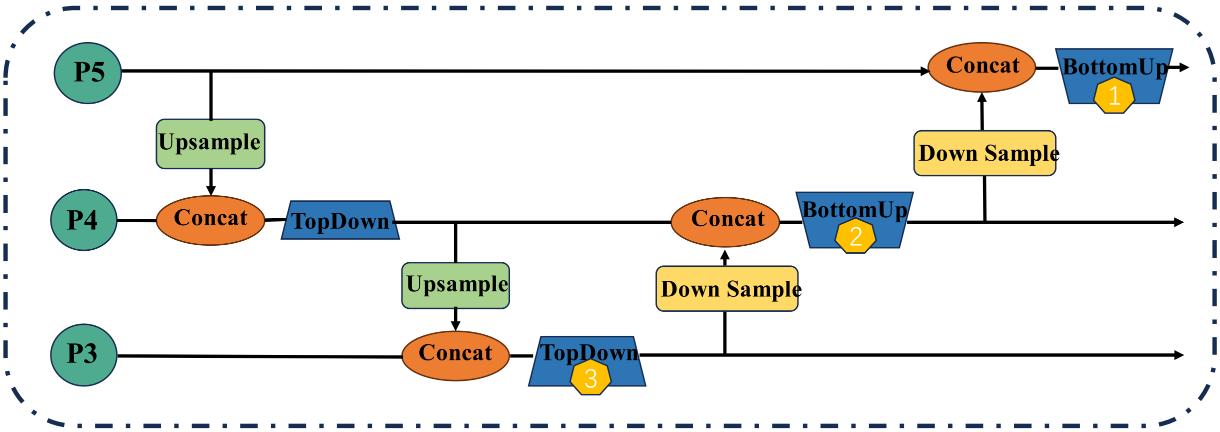


FIGURE 7. The neck sketch of YOLOv5.

Firstly, as illustrated in Figures 5 and 6, the backbone network consists of four Resblock modules, each featuring a Bottleneck architecture. To improve the framework, an ECA block was incorporated into each Resblock. The backbone structure has already proven effective in enhancing network feature identification through both upscaling and downscaling operations. The EB module efficiently extracts defect features using 1D convolutional computation combined with standard convolutional results. The calculation of channel weights is expedited by employing global average pooling, which considers not only the characteristics of individual channels but also those of their k nearest neighbors. Instead of relying on full connectivity for channel downscaling, a one-dimensional convolution is used, resulting in a rapid and efficient process. When the EB module is employed to concurrently augment all four Resblocks, the backbone efficiently executes defect information extraction via both up and downsampling. Moreover, it obtains richer semantic and pictorial information due to the non-degraded convolution, thereby further enhancing its capabilities.

Secondly, the neck network depicted in Figure 5 exhibits a notable enhancement as a result of the fused feature processing by the CC block. The features processed by C3_B2 are then propagated into the CoT module. Initially, the CoT model encodes the contextual data of the input feature keys that have been fused by C3_B2 via a 3×3 convolutional layer. This encoding process produces a static representation of the context of the input features. The encoded feature keys are subsequently appended to the input query, and a dynamic multi-head attention matrix is derived through two successive 1×1 convolutions. The obtained attention matrix is then employed to weigh the input feature values, resulting in a dynamic contextual representation of the input. Ultimately, the dynamic and static contextual representations are merged to produce the final output. Consequently, the CC block consolidates static and dynamic contextual information to alleviate the aliasing effect of top and bottom feature fusions, thereby enhancing performance. We have implemented the refined CC block to adjust the positions of markers 1, 2, and 3 in Figure 7. Based on the experiments detailed in Section D of Chapter V, it has been determined that the modification at marker 1 in Figure 7 is the only improvement to be retained.

IV. EXPERIMENTS

A. EXPERIMENTAL ENVIRONMENT

The operating system Ubuntu version 18.04.6 is used for the experiments in this work. The experiment hardware environments are as follows: graphics processing unit (GPU): NVIDIA GeForce RTX3060/PCle/SSE2 with 12 Gb; central processing unit (CPU): Intel Core i7-7700CPU@3.6GHz $\times$ 8. The experiment software environments are as follows: PyTorch 1.7.0; programming language: Python 3.8. The initial parameters of our method utilized for subsequent experiments are shown in Table 2, and the pre-trained model parameter YOLOv5s.pt was loaded.

TABLE 2. Improved model training parameters.

	Input size	Batch size	Decay	Learning rate	Momentum	Epoch
NEU-DET	320×320	16	0.00005	0.0004	0.8	100
GC10-DET	960×960	16	0.00005	0.0004	0.8	100
SLD-DET	640×640	16	0.00005	0.0004	0.8	100

This experiment uses stochastic gradient descent (SGD), and Cosine Warmup is adopted. The initial learning rate is set to 0.0004, growing linearly in the first 18 epochs and then decreasing in a cosine manner, the weights decay to 0.00005, and the momentum is 0.8. Since the experiments utilize three datasets, the input model size is 320×320 when the image size is 200×200 and 960×960 when the image size is 4096×1000. Due to the dataset SLD-DET varying image sizes, it was chosen to compromise the image input size as 640×640.

B. DATASETS

1) NEU-DET

The NEU-DET dataset, a collection of surface defects in hot-rolled strip steel, was published by Northeastern

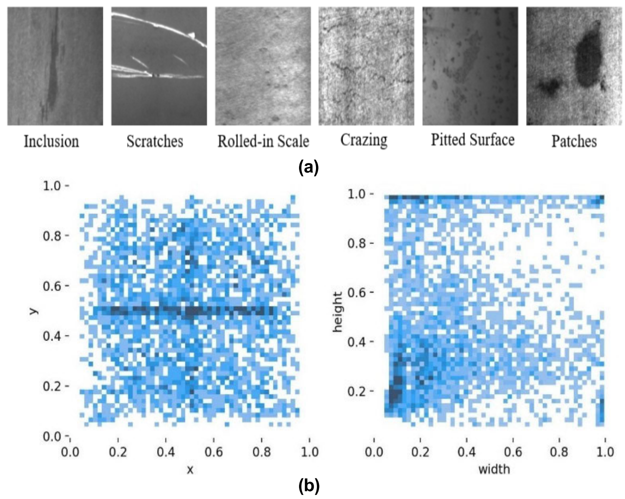


FIGURE 8. (a) NEU-DET dataset and (b) distribution of defects in the NEU-DET dataset.

University [48]. Six common surface defects of strip steel are collected in the dataset: inclusions (In), scratches (Sc), rolled-in scales (Rs), crazing (Cr), pitted surface (PS), and patches (Pa). According to Figure 8(b), most of these defects are small, with widths and heights mainly concentrated between 0.6 × 0.6 and relatively uniform distribution. We randomly separate the defective images from a test set and a training set, as reported by the ratio of 2:8. 1200 images are randomly part from training sets and 300 into test sets.

2) GC10-DET

The GC10-DET data set assembles pictures of surface defects from authentic manufacturing lines. The dataset contains ten common surface defects on the strip line: weld line (Wl), waist folding (Wf), silk spot (Ss), water spot (Ws), oil spot (Os), crescent gap (Cg), inclusions (In), rolled pits (Rp), punching (Pu), and creases (Cr). The dataset comprises 3750 grayscale pictures, with about 200–300 photos of every defect category, and the image size is 4096 × 1000. As evidenced in Figure 9(b), the defect classes in GC10-DET are small, and the width and height distributions are highly uneven. Still, the defects are distributed relatively uniformly in the images. The data set is arbitrarily subdivided into a test set and a training set in proportion to 2:8 for auxiliary experiments.

3) SLD-DET

The SLD-DET dataset is collected by ourselves on the web and in real-life scenarios for steel surfaces containing elongated and small defects. The data set includes 1273 images with defect information for eight defect types: inclusions (In), oil spot (Os), rolled-in scales (Rs), crazing (Cr), scratches (Sc), pitted surface (PS), patches (Pa), and punching (Pu). As evidenced in Figure 10(b), the dataset contains elongated and small defects frequently occurring in steel surfaces. It is arbitrarily subdivided into a test set and a training set in proportion to 2:8 for auxiliary experiments.

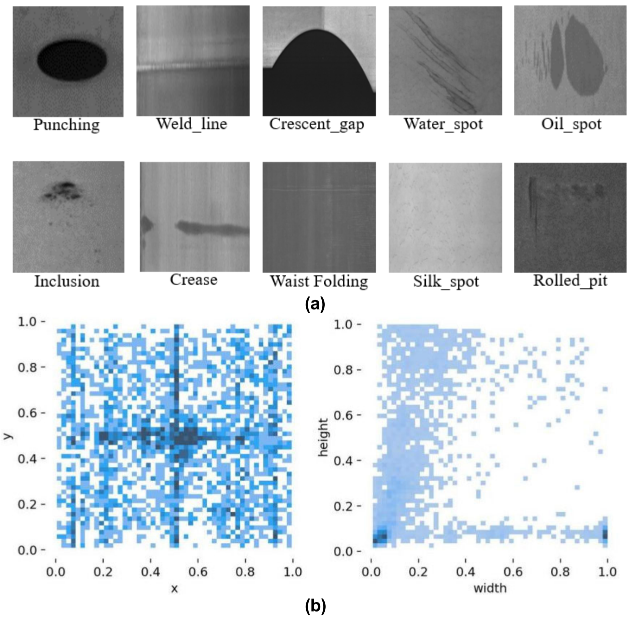


FIGURE 9. (a) GC10-DET dataset and (b) distribution of defects in the GC10-DET dataset.

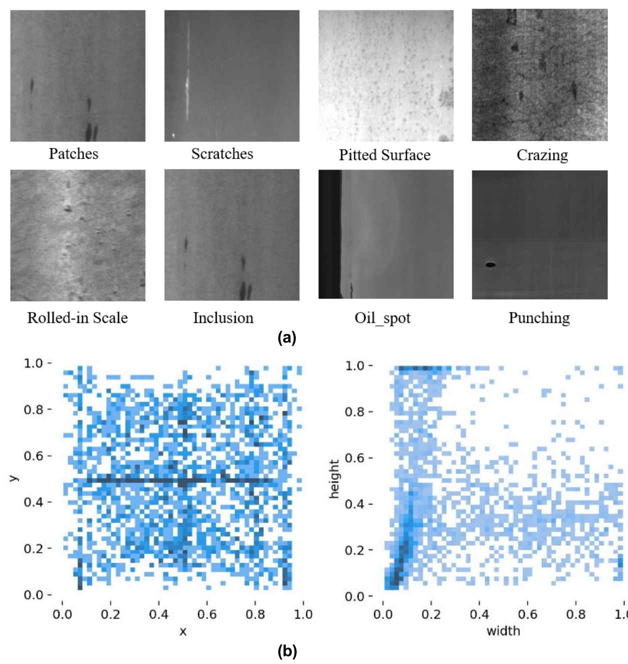


FIGURE 10. (a) SLD-DET dataset and (b) distribution of defects in the SLD-DET dataset.

C. PERFORMANCE EVALUATION

The evaluation metrics involving the target detection results generally include the *Recall*, *Precision*, frames per second (*FPS*), average precision (*AP*), and *mAP*, which are adapted to estimate the instant efficiency and detection method accuracy. We choose the above metrics to measure our results later in our trial. Before presenting these indicators, the following basic concepts are introduced: a true positive (*TP*) suggests that the recognition is a positive case and the recognition is fitting when the correct defect area is detected. A false positive (FP) suggests that this prediction is a positive sample, but the prediction remains false because the wrong defect area is detected. A true negative (TN) suggests that this prediction is a negative sample, and the prediction remains correct when the wrong defect area is not detected. A false negative (FN) suggests that the recognition is a negative case, yet the prediction remains false as the correct defect area is not detected.

Precision: the proportion of true sample correctly determined via the classifier to the total of actual instances determined by the classifier, with the following formula:

$$Precision = \frac{TP}{TP + FP}. \tag{3}$$

Recall: the percentage of the amount of rightly determined true specimens of the classifier to the quantity of all true specimens, and its formula is as follows:

$$Recall = \frac{TP}{TP + FN}. \tag{4}$$

AP: The regions surrounded by precision as the vertical axis and recall as the horizontal axis, with the following formula, where $(r_{i+1} - r_i)$ denotes the horizontal coordinate recall and $P(r_{i+1})$ denotes the vertical coordinate precision:

$$AP = \sum_{i=1}^{n-1} [r_{i+1} - r_i] [P(r_{i+1})]. \tag{5}$$

mAP: in the dataset, the mean value of the AP of total categories is described with the succeeding formula, where m denotes the amount of sample classes within the test set. Where $mAP@0.5{:}0.95$ is the average mAP when the step size is set to 0.05, and the threshold IoU is from 0.5 to 0.95. The $mAP@0.5$ is the mAP if the IoU is 0.5:

$$mAP = \frac{1}{m} \sum_{i=1}^{m} AP_i. \tag{6}$$

FPS: the sum of pictures recognized per second by the target model. The formula for calculating the FPS of the YOLOv5 model is in that manner, where PP indicates the image preprocessing time (which also suggests the inference time), and nms means the non-maximal value suppression processing time in milliseconds:

$$FPS = \frac{1000}{PP + in + nms}. \tag{7}$$

V. RESULTS AND DISCUSSION

A. RESULTS ANALYSIS

Figure 11 indicates the detection performance of the EC-YOLO model upon the NEU-DET data set. Sixteen images were randomly selected and input to the enhanced model simultaneously during the test. A comparison of Figure 11(a) labeling data and (b) prediction data intuitively shows that our proposed approach can accurately identify most defects. The recognition of six defects in the NEU-DET dataset by our improved method is shown in Table 3, including the results of the four categories mAP@0.5, mAP@0.5:0.95, Precision, and Recall. Our method apparently performs well in each aspect of detection, especially for small defects, (In), the mAP reaches 86.7%, and for elongated defects, Sc, it is up to 96.5%. However, for (Cr), the mAP@0.5:0.95 effect is poor at only 20.9%, probably because the Crazing type blends with the background easily, which makes it hard to recognize the defects during detection.

Figure 12 and Table 4 indicate the recognition performance of the EC-YOLO method on the GC10-DET dataset. The Figure 12 (a) labeling data and (b) prediction data intuitively show that most defects can be accurately identified by our proposed method, but there are still some omissions when detecting defects with elongated stripes. Perhaps because of the large variety of defects in the GC10-DET dataset, the contrast between the background and the defects is relatively weak. The recognition of ten defects in

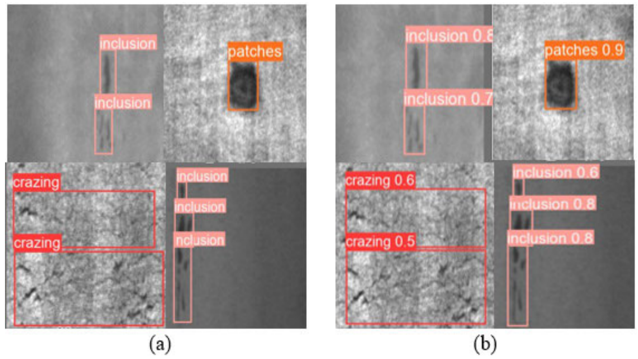


FIGURE 11. Results of testing the NEU-DET dataset compared to labeling results (a)label (b)prediction.

TABLE 3. Performance of the improved method on the NEU-DET dataset.

	Cr	In	Pa	PS	Rs	Sc
Precision	0.676	0.802	0.876	0.867	0.743	0.923
Recall	0.485	0.809	0.934	0.830	0.726	0.917
mAP@0.5	**0.569**	**0.867**	**0.933**	**0.903**	**0.769**	**0.965**
mAP@0.5:0.95	0.209	0.486	0.634	0.538	0.360	0.582

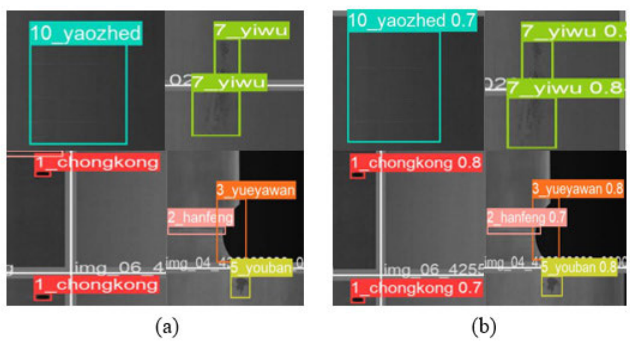


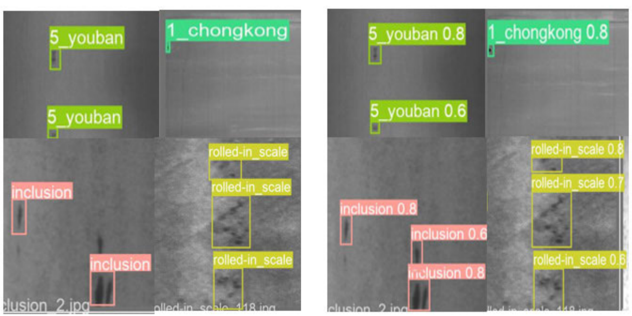
FIGURE 12. Performance of testing the GC10-DET dataset compared to labeling results. (a) label (b)prediction. 1_chongkong for (Pu), 2_hanfeng for (WL), 3_yueyawan for (Cg), 5_youban for (Os), 7_yiwu for (In), and 10_yaozhed for (Wf).

TABLE 4. Performance of the improved method on the GC10-DET dataset.

	Cg	Ws	Pu	Os	Rp	Wf	Wl	Cr	Ss	In
Precision	0.882	0.681	0.810	0.677	0.608	0.621	0.419	0.396	0.727	0.780
Recall	0.934	0.949	1.000	0.817	0.733	0.606	0.477	0.368	0.571	0.727
mAP@0.5	**0.956**	**0.836**	**0.952**	**0.794**	**0.696**	**0.627**	**0.383**	**0.377**	**0.622**	**0.818**
mAP@0.5:0.95	0.577	0.343	0.672	0.447	0.302	0.254	0.153	0.162	0.818	0.334

the GC10-DET dataset by our improved method is shown in Table 4, including the results of the four categories mAP@0.5, mAP@0.5:0.95, Precision, and Recall. Category (In), the mAP was 81.8% for the small defects of concern, but for the elongated defects (Wl), the desired effect was not achieved, only 38.3%. Whereas the overall detection effect of the improved model may meet the industrial requirements, detecting (Cr) and (Wl) defects is still a key research challenge for the later stages of the study.

Figure 13 and Table 5 indicate the recognition performance of the EC-YOLO method on the SLD-DET dataset. It is evident from Figure 13 (a) and (b) that our model is capable of detecting both small and elongated defects, with a high success rate. The recognition of eight defects in the SLD-DET dataset by our improved method is shown in Table 5. The results were very satisfactory for these small and elongated defects, except for the (Cr) category of defects, for which the indicators for all categories were slightly inferior. Especially for (Pa) and (Pu) type defects, the mAP@0.5 can reach 99.3% and 97.8%, which indicates that our model is especially effective for small and elongated defects.



(a) (b)

FIGURE 13. Results of testing the SLD-DET dataset compared to labeling results. (a) label (b)prediction. 1_chongkong for (Pu), 5_youban for (Os).

TABLE 5. Performance of the improved method on the SLD-DET dataset.

	Cr	In	Pa	PS	Rs	Sc	Os	Pu
Precision	1.000	0.887	0.943	0.800	0.842	0.736	0.696	0.916
Recall	0.161	1.000	0.980	0.704	0.941	0.886	0.740	0.956
mAP@0.5	**0.438**	**0.991**	**0.993**	**0.705**	**0.985**	**0.889**	**0.796**	**0.978**
mAP@0.5:0.95	0.176	0.687	0.817	0.436	0.715	0.544	0.374	0.538

B. COMPARISION OF DIFFERENT MODEL

Experiments were designed to compare the efficiency between mainstream object detection algorithms and the improved model on the GC10-DET and NEU-DET datasets. This study compares the results of the EC-YOLO with Faster-RCNN, SSD300 (The size of the input image for SSD300 is 300 $\times$ 300.), SSD512 (The size of the input image for SSD512 is 512 $\times$ 512.), YOLOv3, YOLOv5, YOLOv7, YOLOv8, and YOLOX, where the algorithms are trained and tested using the uniform hardware environment and dataset. The specific performance comparisons for the various experiments are presented in Tables 6 and 7. A comparative visualization of the data for the different indicators is depicted in Figure 14.

TABLE 6. AP and FPS results of the different algorithms on NEU-DET.

Method	backbone	Cr	In	Pa	PS	Rs	Sc	mAP@0.5	FPS
SSD300	VGG16	0.43	0.71	0.90	0.89	0.69	0.63	0.71	82.14
SSD512	VGG16	0.30	0.69	0.90	0.72	0.70	0.65	0.66	17.98
Faster-RCNN	ResNet50	0.48	0.77	0.93	0.83	0.79	0.94	0.79	10.25
YOLOv3	Darknet53	0.23	0.56	0.78	0.79	0.35	0.70	0.57	43.10
YOLOv4	CSPDarkNet53	0.12	0.37	0.41	0.47	0.08	0.16	0.27	34.95
YOLOv7	ELAN-Net	0.19	0.73	0.90	0.76	0.44	0.83	0.64	30.16
YOLOX	CSPDarkNet	0.41	0.80	0.91	0.84	0.75	0.91	0.77	68.92
YOLOv8	CSPDarkNet-C2F	0.52	0.82	0.93	0.87	0.74	0.96	0.81	185.19
YOLOv5	CSPDarkNet53	0.45	0.84	0.92	0.89	0.70	0.95	0.79	91.74
[49]	CSPDarkNet-CBAM	0.51	0.90	0.92	0.83	0.55	0.94	0.78	92.00
[50]	ResNet50	0.76	0.62	0.91	0.90	0.85	0.91	0.82	--
Proposed Method(ours)	CSPDarkNet-EB	**0.57**	**0.87**	**0.93**	**0.90**	**0.77**	**0.97**	**0.83**	**91.74**

TABLE 7. AP and FPS results of the different mainstream algorithms on GC10-DET.

Method	Cg	Ws	Pu	Os	Rp	Wf	Wl	Cr	Ss	In	mAP @0.5	FPS
SSD300	0.89	0.51	0.37	0.23	0.17	0.13	0.10	0.05	0.01	0.01	0.25	71.11
SSD512	0.93	0.92	0.89	0.89	0.64	0.42	0.32	0.23	0.21	0.10	0.55	15.82
Faster-RCNN	0.89	0.68	0.67	0.65	0.52	0.49	0.43	0.37	0.24	0.12	0.51	8.46
YOLOv3	0.89	0.78	0.60	0.50	0.34	0.21	0.20	0.15	0.06	0.00	0.37	39.59
YOLOv4	0.61	0.26	0.16	0.14	0.05	0.01	0.01	0.00	0.00	0.00	0.12	31.92
YOLOv7	0.96	0.93	0.84	0.73	0.53	0.53	0.41	0.41	0.28	0.21	0.58	27.21
YOLOX	0.92	0.82	0.86	0.84	0.75	0.68	0.66	0.55	0.30	0.18	0.67	58.39
YOLOv8	0.95	0.92	0.94	0.80	0.70	0.59	0.39	0.27	0.57	0.72	0.68	14.66
YOLOv5	0.94	0.87	0.93	0.78	0.72	0.64	0.35	0.33	0.48	0.68	0.67	85.47
Proposed Method(ours)	**0.96**	**0.84**	**0.95**	**0.79**	**0.70**	**0.63**	**0.38**	**0.38**	**0.62**	**0.82**	**0.71**	**64.10**

By comparison, based on the various experimental results of Table 6, Table 7, and Figure 14, the following inferences can be drawn:

- The initial ten rows in Table 6 demonstrate the quantitative outcomes of comparing our methodology to the traditional target detection method on the NEU-DET dataset. Whether comparing the benchmark model or the mainstream target detection model, the detection performances of our approach are compelling. Specifically, our method achieves

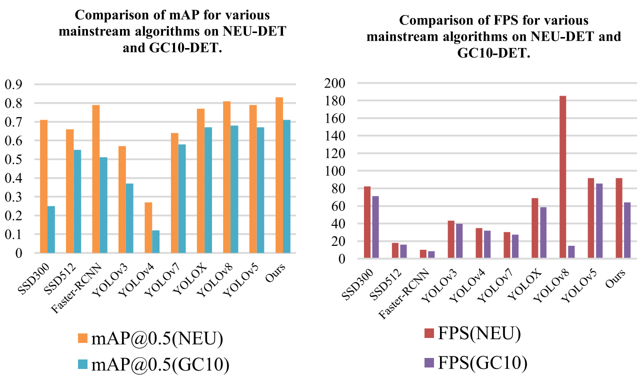


FIGURE 14. **The performance of mainstream detection models tested on NEU-DET and GC10-DET datasets.**

a mAP@0.5 of 83% and a FPS of 91.74, which surpasses the average accuracy of the baseline model by four percentage points, thus achieving a balance between detection speed and accuracy. Our method demonstrates superior performance, particularly when addressing small defects (In) and elongated defects (Sc). It surpasses all mainstream models, achieving a 3% improvement for small defects and a 1% improvement for elongated defects over the best-performing results in these respective categories. Table 6, specifically rows 11 and 12, demonstrates that our method achieves a superior balance between detection accuracy and speed on the NEU-DET dataset compared to other methods focused on identifying minor defects in steel surfaces.

- Table 7 displays the quantitative results of comparing the conventional target detection method with our proposed approach on the GC10-DET dataset. As can be observed, certain models within the GC10-DET dataset exhibit subpar detection outcomes. Nevertheless, our enhanced model continues to fulfill the industrial requirements for real-time inspection, achieving a mAP@0.5 and a FPS rate of 71% and 64.10, respectively. Although our method does not detect elongated defects (Wl) well, only 38%, it still outperforms most of the mainstream models, and it achieves 82% detection of small defects (In), which makes it very encouraging. It is analyzed that the huge inter-class gap and slight intra-class gap of defects in the GC10-DET dataset and the inferior contrast between the defect features and the ground may be the direct factors for the inferior detection results. And even though the backbone network fully utilizes channel attention whereas the necking network adequately fuses semantic and localization features to avoid aliasing effects, the defects (Wl)and (Cr) fail to resist these interferences.
- Figure 14 presents an intuitive visualization of the mAP@0.5 and FPS test performance for each mainstream detection model on NEU-DET and GC10-DET datasets. For detection accuracy, the diagram presented in Figure 14(a) illustrates that the accuracy of our improved method outperforms other mainstream models on either dataset. The graph in Figure 14(b) exhibits the detection speed, and our test speed is superior to other mainstream models on the NEU-DET dataset. On the GC10-DET dataset, our test speed significantly lags behind the state-of-the-art model YOLOv8. However, despite its limitations, it still meets the requirements for real-time detection. Compared to the benchmark model, our method demonstrates consistent recognition speed on the NEU-DET dataset. However, there is a 21.37% decrease in speed on the GC10-DET dataset. This discrepancy can be attributed to the differing input image sizes: the former has an input size of 320 × 320, whereas the input size of latter is 960 × 960. The larger input size results in increased elapsed time during image preprocessing and inference.

C. ABLATION STUDY

To validate that a single improvement block positively impacts the original model, we designed ablation trials to compare the effectiveness of the improvement points. The ablation experiments are implemented on two publicly available surface defect datasets, GC10-DET and NEU-DET. To further understand the impact of the two elevation modules on detecting small and elongated defects, ablation experiments were performed simultaneously on the SLD-DET dataset.

TABLE 8. **Specific performance of the ablation trials on three datasets.**

Datasets		EB block	CC block	mAP@0.5	Parameters (MB)
NEU-DET	baseline			0.790	702.63
		√		0.820	702.63
			√	0.811	819.44
	Ours	√	√	0.834	819.44
GC10-DET	baseline			0.671	703.71
		√		0.687	703.71
			√	0.699	820.55
	Ours	√	√	0.706	820.52
SLD-DET	baseline			0.847	703.17
		√		0.868	703.17
			√	0.868	819.99
	Ours	√	√	0.875	819.99

To validate the effectiveness of the EB, an ablation study was performed using the original YOLOv5 approach as a reference. Using the EB block, we have modified the p2, p3, p4, and p5 modules in the Figure 6 backbone. The specified performances are shown in rows 3, 7, and 11 of Table 8. In contrast to the original YOLOv5 method, the model accuracy is enhanced by 3% on the NEU-DET, 1.6% on the GC10-DET, and 2.8 % on the SLD-DET. After we adopted the EB block to modify the backbone, the quantity of parameters of all three models remained equal to the baseline method. Based on this analysis, it can be deduced that the EB generates a more comprehensive feature information through cross-channel interaction. The non-degradation operation guarantees that the network parameters remain constant.

To verify the effectiveness of the CC block, we modify the YOLOv5 neck network PAFPN by using the CC block. In Figure 7, the neck network in Part 1 uses a CC block for improvement. According to the ablation experimental results in rows 4, 8, and 12 of Table 8, it can be concluded that the CC block placed at part 1, where the feature fusion yields an improvement of 2.1%, 2.8%, and 2.8% in the NEU-DET, GC10-DET, and SLD-DET datasets, respectively. Thus, the improved CC module combined with the contextual capability can benefit network feature fusion. Adding the CC module to the neck network increases the parameters of our model by around 116.81MB in the three comparison sets. This modification improves accuracy by integrating contextual information and reducing the impact of overlapping. However, the larger model size may slow down the detection efficiency.

In rows 5, 9, and 13 of Table 8, the test results on three different test sets are exhibited when the EB and the CC blocks work on both the backbone and the neck network of the benchmark. The test performances on the NEU-DET, GC10-DET, and SLD-DET datasets are 83.4%, 70.6%, and 87.5%, respectively, which are 4.4%, 3.5%, and 2.8% higher than the benchmark method results. All the data indicate that non-dimensionalized channel attention allows the backbone network to collaborate effectively with the necking network, which concentrates on contextual information, to perform the defect detection task with greater precision.

D. EXPLORATION OF SMALL AND ELONGATED DEFECT DETECTION

To explore the effect of attention utilization during feature extraction and feature fusion for small and elongated defects and understand whether these defects are more sensitive to top-level semantic or bottom-level positional information, the following exploratory experiments are continued with the SLD-DET dataset. Firstly, we utilize the EB block to improve the Resblock p3, p4, or p5 in Figure 6. We set the module EB with channel attention on p3, p4, or p5 to observe how the detection result changes and then analyze the efficiency of the operation. The test results are demonstrated in Table 9. We conducted experiments utilizing the CC module for the feature fusion scales labeled 1, 2, and 3 in Figure. 7, respectively, and the test results are demonstrated in Table 10.

TABLE 9. AP performance of the different positions of the EB.

Datasets		P3	P4	P5	mAP
SLD-DET	EB block	√			0.855
			√		0.867
				√	0.873

TABLE 10. AP performance of the different positions of the CC.

Datasets		Outpu1	Output2	Output3	mAP
SLD-DET	CC block	√			0.840
			√		0.860
				√	0.868

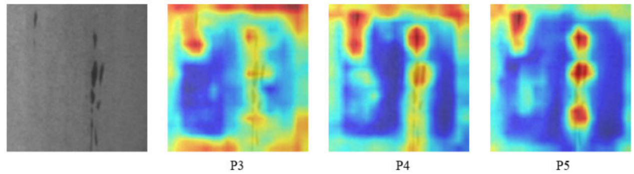


FIGURE 15. Characteristic visualization of EB action at different scales.

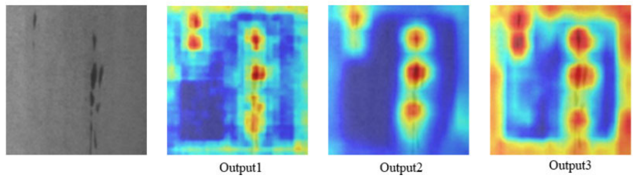


FIGURE 16. Characteristic visualization of CC action at different scales.

As exhibited in Figure. 6, as the degree of convolution of p3, p4, and p5 is deepened sequentially, the extracted semantic data of the defective features in the backbone network is richer. In comparison, the graphical information is weaker, e.g., compared with p3 and p5, the graphical data of the extracted defective features in the p3 module should be stronger than the extracted ones in the p5 module, and the semantic information should be weaker than the extracted ones in the p5 module. As displayed in Table 9, the defect detection accuracy gradually increases when we utilize the EB module to modify p3, p4, and p5 individually. It can be inferred that when using the non-decreasing channel attention module EB for feature extraction of small and elongated steel surface defects, it is more reasonable to trade off semantic and graphical information; the former assists the model in improving the accuracy of defect detection, and the latter assists the model to localize the defects better. We output the thermal detection map at improved p3, p4, and p5, as shown in Figure 15. This reveals that the non-dimensionalized channel attention block EB allows the model to concentrate on the defect object more precisely.

Figure 7 shows that the feature fusion network with an FPN-based structure obtains more comprehensive defect features after successive up-and-down sampling. The labeled 1, 2, and 3 modules indicate the feature fusion blocks when the input images are downsampled 32 times, 16 times, and 8 times, respectively. Comparing inputs 1 and 3, the former is downsampled to a high degree. It contains enriched semantic information, whereas the latter is downsampled to a relatively lower degree and holds enriched graphical information. As can be seen from Table 10, there is a noticeable and continuous increase in defect detection accuracy when we utilize the context-focused CC module to optimize the output 1, 2, and 3 levels. Therefore, it can be hypothesized that small and elongated defects on the surface of steel may be more sensitive to high-level semantic information during feature fusion. The CC module, which pays more attention to context, can better equilibrate the fusion of semantic and graphical information to improve defect detection accuracy. We output the thermograms of detection with enhanced output headers 1, 2, and 3, as shown in Figure 16. It exhibits that the CC module focusing on the global context allows the model to focus more accurately on defect information, expands the scope of attention, and enhances the model feature linkage capability.

As we all know, small and elongated defects detection generally concentrate more on extracting graphical information. However, it has been observed that the detection of such defects on steel surfaces is particularly reliant on semantic information during both feature extraction and fusion. The non-decreasing channel attention strengthens the attention of the backbone to these defects. The self-attention in connection with context alleviates the aliasing effects of the convolutional network.

VI. CONCLUSION

The downstream steel industry has high expectations and confidence in steel quality. In our study, an enhanced YOLOv5 algorithm, EC-YOLO, is designed to tackle the problem of high-precision identification of surface defects on strips via enhanced recognition of small and elongated defects. First, the bottleneck block of the original YOLOv5 backbone network was improved by utilizing the ECA block, where the EB block was proposed to strengthen the capability of the backbone network to derive channel features. Then, the CC block was utilized to enhance feature fusion in the original YOLOv5 algorithm. Furthermore, we conducted experiments on the SLD-DET dataset, which encompasses both small and elongated defects, to investigate whether these defects exhibit heightened sensitivity to top-level semantic knowledge during feature extraction and fusion. The experimental results indicated that our enhanced approach can accurately and swiftly identify various defects within the dataset. This methodology offers an innovative solution for the detection of defects on strip surfaces. Nevertheless, the increased parameters in the proposed method complicate our model. Thus, it will continue to be optimized according to these experimental results and problems.

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